

Recurrence Analysis of Integrally Private Support Vector Machine

Ayush K. Varshney

Vicenç Torra

Department of computing science, Umeå University, Sweden

AYUSHKV@CS.UMU.SE

VTORRA@CS.UMU.SE

Abstract

Integral privacy, an alternative to k-Anonymity and differential privacy, focuses on creating ambiguity for intruders by considering models generated from diverse datasets as privacy-preserving. Integral privacy calls such models as recurring models. While prior research has primarily explored recurrence in deep learning models which have large parameter space, this paper addresses the recurrence analysis of a typical machine learning model with relatively small parameter space like Support Vector Machine (SVM). Models having small parameter space can have significant impact due to the presence and absence of a datapoint. Due to this reason, one may intuitively consider that their probability to recur is low. We challenge this hypothesis with the recurrence analysis of SVM models trained with mean samplers like stochastic gradient descent. We find that under constrained environment SVM models recurs with high probability. This research enhances our understanding of privacy-preserving models in the context of SVMs, providing valuable insights into their privacy guarantees.

Keywords: Machine learning; Support Vector Machine; Recurrence Analysis; Integral privacy.

1. Introduction

In the era of big data, the Machine Learning (ML) applications are increasingly embedded in daily life, leveraging large datasets to enhance model performance. However, this data abundance raises privacy concerns, prompting regulations like GDPR (Voigt and Von dem Bussche (2017)) and CCPA (Goldman (2020)) to ensure responsible data handling. Traditional privacy-preserving techniques, such as k-Anonymity Samarati (2001); Samarati and Sweeney (1998) and differential privacy Dwork (2006), offer some protection but face limitations, especially in applications requiring high data utility.

Integral privacy Torra et al. (2020) was proposed as an alternative to both k-Anonymity and Differential Privacy (DP). It defines a model as privacy-preserving if it can be generated by multiple disjoint training datasets, introducing ambiguity for intruders attempting membership inference and model comparison attacks. Specifically, the set of integrally private models consists of recurring models that can be generated with multiple datasets. However, obtaining diverse datasets for a given application can be challenging.

Senavirathne et al. Senavirathne and Torra (2019) proposed a method to identify integrally private decision trees by estimating model recurrence with an approximated model space. While suitable for small datasets, this method becomes computationally expen-

sive for larger datasets and is even more challenging for deep learning models due to their huge parameter space. To address this, Varshney and Torra (2023) proposed a relaxed version, Δ -Integral Privacy, showing that similar training subsamples produce similar models. This finding aligns with the observations made by Thudi et al. (2022) and Varshney and Torra (2024), provides probabilistic recurrence guarantees, though primarily focused on deep learning models with extensive parameters.

However, effect of a single minibatch on a weight in deep learning models is often minimal, as adjustments are distributed across many parameters. In contrast, a minibatch's impact on a simpler model like Support Vector Machine (SVM) Cortes (1995) is more significant due to fewer parameters in SVM. Similar to Varshney and Torra (2023), Kwatra et al. (2023) gives the methodology to generate recurring SVM models. First, the methodology randomly selects disjoint subsamples from the training data. Then, it trains the SVM models on each subsample. The trained SVM models are compared to find a set of recurring models. The requirement of disjoint subsamples is required in order to avoid any membership inference analysis. The methodology provides a set of recurring models and their statistics but lacks theoretical guarantees on the recurrence. Hence, in this paper, we present the recurrence analysis for SVM models and show that with high probability, a typical machine learning model like SVM also recurs.

2. Recurrence Analysis SVM

SVM, or Support Vector Machine, is a widely recognized binary classifier. Given a dataset $D_{n \times d}$, SVM identifies the most effective hyperplane from a set of hyperplanes to differentiate between data samples belonging to distinct classes. The optimal SVM hyperplane satisfies $J(w, b): w^T x + b = 0$, where $w \in \mathcal{R}^n$ is the normal vector of $J(w, b)$, $b \in \mathcal{R}$ represents the bias term. The optimal hyperplane is solved using the following optimization problem.

$$J(w, b) : \min_{w \in \mathbb{R}^n} \frac{1}{2} w^T w + C \sum_{i=1}^N \max(0, 1 - y_i(w^T x_i + b))$$

$$\text{subject to, } y_i(w^T x_i + b) \geq 1 - \max(0, 1 - y_i(w^T x_i + b)), \text{ for } i = 1, 2, \dots, d \quad (1)$$

Here, $y_i \in \{-1, 1\}$ denotes the label of the i^{th} sample and the parameter C regulates the trade-off between maximizing the margin and minimizing number of incorrectly classified samples. The typical update rule with mean samplers (refer Appendix A for mean samplers) is given with $w^t \leftarrow w^{t-1} - \eta \nabla J^t(w^{t-1})$.

Then, from the objective function in Eq. (1)

$$\nabla J^t(w^{t-1}) = \begin{cases} w^{t-1} - C \sum_i y_i x_i, & \text{if } 1 - y_i(w^T x_i + b) \geq 0 \\ w^{t-1}, & \text{otherwise} \end{cases}$$

if $y_i w^T x_i \leq 1$ then,

$$w^t \leftarrow (1 - \eta) w^{t-1} - \eta C \sum_i y_i x_i \quad (2)$$

$$\text{otherwise, } w^t \leftarrow (1 - \eta) w^{t-1} \quad (3)$$

Now, let us consider a set of data samples, $D_1, D_2, \dots, D_m$, i.i.d. (independent and identically distributed) sampled from a given dataset $\mathcal{D}$ with some distribution and N number of data points. $M_1, M_2, \dots, M_m$ be the SVM models we want to train on them. Since, we know from Algorithm 1 that all the models are initialized with 0 ($\in \mathcal{R}^n$). From Eq. (2) and (3), we can say that for two different models having the same learning rate (η), C , and N , the weights in the 1st iteration will depend on the product of $y_i x_i$. Similarly, if two models have the same weights at some iteration $t - 1$ then at iteration t their weights will depend on the product of $y_i x_i$ sampled at iteration t .

Let each of the $\mathcal{D}_i$ be composed of b minibatches $(x_1, x_2, \dots, x_b)$. Since, the data samples are chosen iid from $\mathcal{D}$, samples from these data samples will also follow the similar distribution. Let the mean of the product $y_i x_i$ for each of the minibatch at any iteration be μ and the trace of the covariance matrix be σ^2 ($= \sum_{i=1}^n \sigma_i^2$, where σ_i^2 is the covariance of the i^{th} component of a random variable x sampled with distribution $\mathcal{D}$) as the trace of the covariance matrix. Note here that mean value of $y_i x_i$ for all minibatch would still be μ ($= \frac{1}{b} \sum_{i=1}^b \mu$) but mean sampling of trace of the covariance matrix will be $\frac{1}{b} \sigma^2$. Then by Markov's inequality we can say that for a given Δ ,

$$P(|\frac{1}{N} \sum y_i x_i - \mu|_2 \geq \Delta) = P(|\frac{1}{N} \sum y_i x_i - \mu|_2^2 \geq \Delta^2) \leq \frac{E(|\frac{1}{N} \sum y_i x_i - \mu|_2^2)}{\Delta^2}$$

Here the first equality is true by the property of monotonicity of squares and second is Markov's inequality. Note that $E(|\frac{1}{N} \sum y_i x_i - \mu|_2^2)$ is just the trace of the covariance matrix ($\frac{1}{b} \sigma^2$), then we can write:

$$P(|\frac{1}{N} \sum y_i x_i - \mu|_2^2 \geq \Delta^2) \leq \frac{\sigma^2}{b\Delta^2}$$

From this equation, we can infer the following (as $P(a < b) + p(a \geq b) = 1$)

$$P(|\frac{1}{N} \sum y_i x_i - \mu|_2^2 \leq \Delta^2) \geq 1 - \frac{\sigma^2}{b\Delta^2}$$

Let us consider two models M_j and M_k . Then, let us consider the triangle inequality ($|a - b| \leq |a| + |b|$) in relation to the data used to train these models. Then, if $|\frac{1}{N} \sum y_i^j x_i^j - \mu|_2^2 \leq \Delta^2$ with probability at least $\kappa = 1 - \frac{\sigma^2}{b\Delta^2}$ and $|\frac{1}{N} \sum y_i^k x_i^k - \mu|_2^2 \leq \Delta^2$ with probability κ , it follows that $|\frac{1}{N} \sum y_i^j x_i^j - \frac{1}{N} \sum y_i^k x_i^k|_2 \leq 2\Delta^2$ with probability at least κ^2 (assuming independence in the computations of the models).

Then, the probability that among the m different models, there exists two of them that are in the Δ^2 ball of μ is at least:

$$\sum_{r=2}^m \binom{m}{r} \left(1 - \frac{\sigma^2}{b\Delta^2}\right)^r \left(\frac{\sigma^2}{b\Delta^2}\right)^{m-r} \quad (4)$$

Similarly, the probability that k models out of m are in the Δ ball of μ would be:

$$\left(1 - \frac{\sigma^2}{b\Delta^2}\right)^m \quad (5)$$

Equation (4) and (5) represents the probability of weights updating in the Δ^2 ball of μ for a single iteration. For T iterations, the probability that there exists at least two models having weight updates in the Δ^2 ball of μ is $\geq (\sum_{r=2}^m \binom{m}{r} \left(1 - \frac{\sigma^2}{b\Delta^2}\right)^r \left(\frac{\sigma^2}{b\Delta^2}\right)^{m-r})^T$. Similarly, after T iterations, the probability of all m models to be in the Δ^2 ball of μ is $\geq \left((1 - \frac{\sigma^2}{b\Delta^2})^m\right)^T$.

So, for data samples sampled iid from some dataset, we can conclude that the lower bound for the probability that there exists recurrent models within the Δ^2 ball is at least $\sum_{r=2}^m \binom{m}{r} \left(1 - \frac{\sigma^2}{b\Delta^2}\right)^r \left(\frac{\sigma^2}{b\Delta^2}\right)^{m-r}$, and that all the m models are in the Δ^2 ball of μ is at least $((1 - \frac{\sigma^2}{b\Delta^2})^m)^T$.

We establish this result in the following theorem.

Theorem 1 *Let $D_1, D_2, \dots, D_m$ be i.i.d. (independent and identically distributed) samples from a given dataset $\mathcal{D}$, let $\Delta > 0$, and let $M_1, M_2, \dots, M_m$ represent SVM models built from data $D_1, \dots, D_m$ using Algorithm 1 for a given number of iterations T . Then, the probability of having at least two recurrent models within the Δ^2 ball is at least $\sum_{r=2}^m \binom{m}{r} \left(1 - \frac{\sigma^2}{b\Delta^2}\right)^r \left(\frac{\sigma^2}{b\Delta^2}\right)^{m-r}$, and the probability that all m models are in the Δ^2 ball of μ is at least $((1 - \frac{\sigma^2}{b\Delta^2})^m)^T$.*

Remark on the choice of Δ, b, T, m : To increase the likelihood of achieving recurrent models, several factors should be considered. Firstly, increasing the number of i.i.d samples m and Δ (the error) value positively correlates with the recurrence of models. Furthermore, reducing the number of batches, b , effectively increases batch sizes, and low values of T (avoid overtuning) is also beneficial for higher likelihood.

In order to generate higher recurring models, we can say that increasing the number of i.i.d samples (m), low number of batches b (higher batch sizes), low T (avoid overtuning) and high Δ (the error) value increases the probability of getting recurrent models.

The experimental results in Kwatra et al. (2023) show the high recurrence of SVM for five dataset with varying sizes ($\approx 54k$ -2M instances). The results show that the integrally private models obtain good results in accuracy, recall, precision, and F1 score compared to the baseline SVM model and its differentially private counterpart (DP-SVM) model.

3. Discussion and Conclusion

In this paper, we present the recurrence analysis for integrally private Support Vector Machines (SVMs). The integral privacy model selects models which recurs from multiple datasets, introducing uncertainty for potential intruders. Unlike existing privacy models in the literature, such as k-Anonymity and ϵ -Differential Privacy (ϵ -DP), which primarily focus on preserving the privacy of stored data or the privacy during model inferences, integral privacy centers around the selection of private models during training. Unlike k-Anonymity and ϵ -DP, integrally private models does not cost much utility. The privacy budget in integral privacy depends on the Δ value (see Eq. (5)). A higher value of Δ means higher difference in the models and reduced privacy.

The analysis presented in this paper emphasizes that when mean samplers, like the SGD optimizer, are employed during training, models tend to exhibit recurrence with a high

probability, regardless of the data type or model size. However, certain machine learning models, such as decision trees and k-nearest neighbors, do not utilize mean samplers in their training process. Further recurrence analyses are needed to confirm whether machine learning models that do not employ mean samplers also exhibit recurrence.

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Appendix A. Mean Samplers

SVM are the supervised machine learning models which train on the labeled dataset $\mathcal{D}$. Usually, we want to learn weights and biases (say w) of a model M using training data X ($\subseteq \mathcal{D}$) so that it returns the class label ($\{1, 2, \dots, c\}$) for any given instance. The training of the model M is done iteratively from the i.i.d. sampled n batches. Each batch consists of a set of records (x_i, y_i) . In each epoch, for parameters w in model M the loss $\mathcal{L} : M_w \times y \rightarrow [0, \infty)$ computes the classification error for the batch. Iteratively, we aim to minimize the $\mathcal{L}$ and update the parameters using some update rule g i.e. $w_{t+1} \leftarrow g(w_t, X)$. Mean samplers are an important class of update rules, where the training dataset X is divided into n batches i.e. $X = \bigcup_{i=1}^n \hat{x}_i$, $\hat{x}_i \cap \hat{x}_j = \emptyset$ and each batch consists of b data instances $\hat{x}_i = \{x_1, x_2, \dots, x_b\}$. Here, the update rule looks like: $w_{t+1} \leftarrow g(w_t, \hat{x}_i) = \frac{1}{b} \sum_{i=1}^b g(w_t, (x_i, y_i))$. Mean samplers like minibatch stochastic gradient descent (SGD) samples a minibatch and update the parameters to minimize the average loss for the minibatch. This variant of SGD is known for its applicability in machine learning algorithms. Here, a typical update rule looks like:

$$w_{t+1} = g(w_t, \hat{x}_k) = w_t - \eta \frac{1}{b} \sum_{i=1}^b \nabla_w \mathcal{L}(M_w(\hat{x}_i, y_i))|_{w_t} \quad (6)$$

Where η is the learning rate and $\frac{1}{b} \sum_{i=1}^b \nabla_w$ is the average gradient with respect to w_t for $\hat{x}_i$ minibatch. A set of models is considered similar if, for each model, the average gradients learned from their respective minibatches are the similar. We use this analogy to find the probability of recurrence of models.

Appendix B. Support Vector Machine

In the objective function of SVM (refer Eq. (1)), minimizing the L_2 norm regularization is equivalent to maximizing the margin in the first term. The second term is to penalize incorrectly classified samples. The initial set of constraints ensures that the projections of data points are separated by at least one unit. If this condition can not be satisfied, minimizing the error variable leads to the formation of soft-margin hyperplane. A typical algorithm for learning the weights of SVM is given in Algorithm 1.

Algorithm 1: Mini-Batch Gradient Descent for SVM

Input : S - Training set, M - Minibatch size, T - Number of epochs, η - Learning rate
Output: w - Learned weights

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Initialize  $w \leftarrow 0 \in \mathbb{R}^n$  for  $t = 1$  to  $T$  do
  Randomly shuffle the training set  $S$  for each minibatch  $B \subseteq S$  of size  $M$  do
    Compute gradient estimate:  $\nabla J_B(w) = \frac{1}{M} \sum_{(x_i, y_i) \in B} \nabla J(w; x_i, y_i)$ 
    Update weights:  $w \leftarrow w - \eta \nabla J_B(w)$ 
  end
end
return  $w$ 

```
